DESIGN CODE OF FULL ADDER (VERILOG)

src/fa_verilog.v [Verilog]

```verilog
`timescale 1ns / 1ps
//////////////////////////////////////////////////////////////////////////////////
// Company:
// Engineer:
//
// Create Date: 10/01/2026 04:43:21 PM
// Design Name:
// Module Name: fa_verilog
// Project Name:
// Target Devices:
// Tool Versions:
// Description:
//
// Dependencies:
//
// Revision:
// Revision 0.01 - File Created
// Additional Comments:
//
//////////////////////////////////////////////////////////////////////////////////


module fa_verilog(

    input a,
    input b,
    input cin,
    output s,
    output cout

    );

        assign s = a^b^cin;

        assign cout = (a&b)|(b&cin)|(cin&a);

endmodule
```

DESIGN CODE OF 4-BIT RIPPLE CARRY ADDER (VERILOG)

src/rca4_verilog.v [Verilog]

```verilog
`timescale 1ns / 1ps
//////////////////////////////////////////////////////////////////////////////////
// Company:
// Engineer:
```

```
//
// Create Date: 10/01/2026 04:42:48 PM
// Design Name:
// Module Name: rca4_verilog
// Project Name:
// Target Devices:
// Tool Versions:
// Description:
//
// Dependencies:
//
// Revision:
// Revision 0.01 - File Created
// Additional Comments:
//
//////////////////////////////////////////////////////////////////////////////////


module rca4_verilog(
        input [3:0] a,b,
        input cin,
        output [3:0] s,
        output cout
    );

    wire [4:0] c;

    assign c[0] = cin;
    assign cout = c[4];

    genvar i;

    generate

        for(i = 0 ; i < 4 ; i = i + 1) begin : fa

            fa_verilog fa (a[i] , b[i] , c[i] , s[i] , c[i+1]);

        end


    endgenerate

endmodule
```

TESTBENCH CODE OF 4-BIT RIPPLE CARRY ADDER (VERILOG)

src/rca4_tb.v [Verilog]

```
`timescale 1ns / 1ps
```

```
//////////////////////////////////////////////////////////////////////////////////
// Company:
// Engineer:
//
// Create Date: 10/01/2026 05:04:36 PM
// Design Name:
// Module Name: rca4_tb
// Project Name:
// Target Devices:
// Tool Versions:
// Description:
//
// Dependencies:
//
// Revision:
// Revision 0.01 - File Created
// Additional Comments:
//
//////////////////////////////////////////////////////////////////////////////////


module rca4_tb();

    reg [3:0] a , b;
    reg cin;
    wire [3:0] s;
    wire cout;
    integer i , j ,k;

    rca4_verilog rca4_test (a , b , cin , s , cout);

    initial begin

            for (i = 0 ; i < 16 ; i = i + 1) begin
                for (j = 0 ; j < 16 ; j = j + 1) begin
                    for (k = 0 ; k < 2 ; k = k + 1) begin

                        a <= i;
                        b <= j;
                        cin <= k;

                        #10;

                    end
                end
            end
 end

endmodule
```

CONSTRAINT FILE FOR BASYS 3 BOARD (XDC)

src/rca4.xdc [XDC]

```
## This file is a general .xdc for the Basys3 rev B board
## To use it in a project:
## - uncomment the lines corresponding to used pins
## - rename the used ports (in each line, after get_ports) according to the top level signal names in the project

## Clock signal
#set_property -dict { PACKAGE_PIN W5   IOSTANDARD LVCMOS33 } [get_ports clk]
#create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports clk]

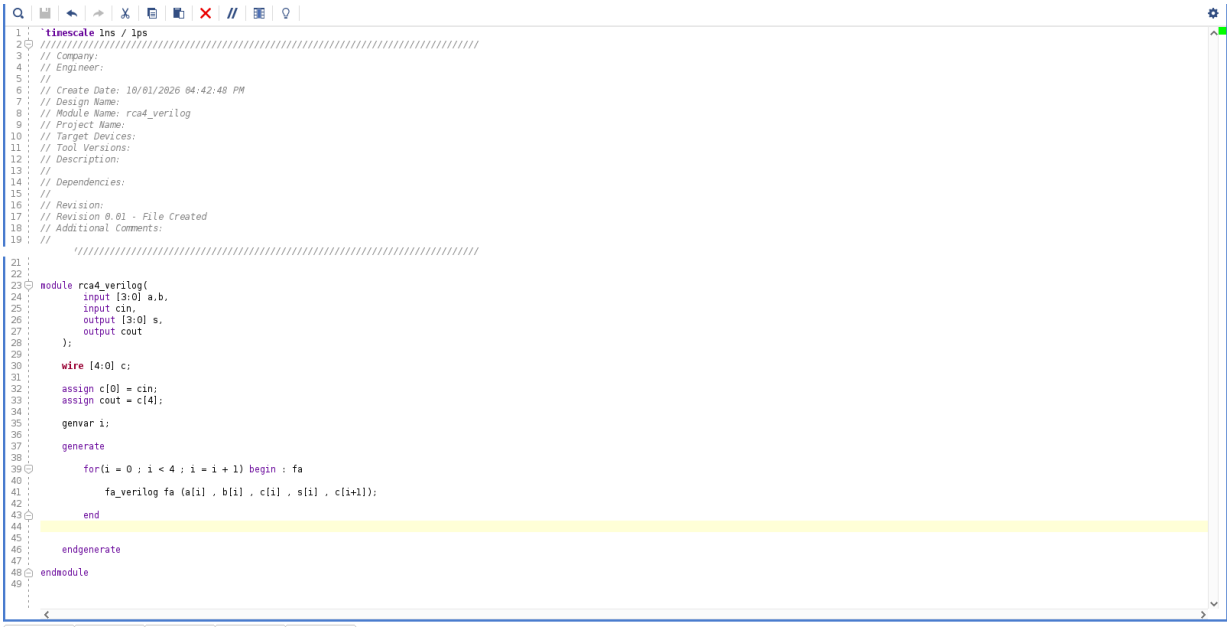

## Switches
set_property -dict { PACKAGE_PIN V17   IOSTANDARD LVCMOS33 } [get_ports {a[0]}]
set_property -dict { PACKAGE_PIN V16   IOSTANDARD LVCMOS33 } [get_ports {a[1]}]
set_property -dict { PACKAGE_PIN W16   IOSTANDARD LVCMOS33 } [get_ports {a[2]}]
set_property -dict { PACKAGE_PIN W17   IOSTANDARD LVCMOS33 } [get_ports {a[3]}]
set_property -dict { PACKAGE_PIN W15   IOSTANDARD LVCMOS33 } [get_ports {b[0]}]
set_property -dict { PACKAGE_PIN V15   IOSTANDARD LVCMOS33 } [get_ports {b[1]}]
set_property -dict { PACKAGE_PIN W14   IOSTANDARD LVCMOS33 } [get_ports {b[2]}]
set_property -dict { PACKAGE_PIN W13   IOSTANDARD LVCMOS33 } [get_ports {b[3]}]
set_property -dict { PACKAGE_PIN V2    IOSTANDARD LVCMOS33 } [get_ports {cin}]
#set_property -dict { PACKAGE_PIN T3    IOSTANDARD LVCMOS33 } [get_ports {sw[9]}]
#set_property -dict { PACKAGE_PIN T2    IOSTANDARD LVCMOS33 } [get_ports {sw[10]}]
#set_property -dict { PACKAGE_PIN R3    IOSTANDARD LVCMOS33 } [get_ports {sw[11]}]
#set_property -dict { PACKAGE_PIN W2    IOSTANDARD LVCMOS33 } [get_ports {sw[12]}]
#set_property -dict { PACKAGE_PIN U1    IOSTANDARD LVCMOS33 } [get_ports {sw[13]}]
#set_property -dict { PACKAGE_PIN T1    IOSTANDARD LVCMOS33 } [get_ports {sw[14]}]
#set_property -dict { PACKAGE_PIN R2    IOSTANDARD LVCMOS33 } [get_ports {sw[15]}]


## LEDs
set_property -dict { PACKAGE_PIN U16   IOSTANDARD LVCMOS33 } [get_ports {s[0]}]
set_property -dict { PACKAGE_PIN E19   IOSTANDARD LVCMOS33 } [get_ports {s[1]}]
set_property -dict { PACKAGE_PIN U19   IOSTANDARD LVCMOS33 } [get_ports {s[2]}]
set_property -dict { PACKAGE_PIN V19   IOSTANDARD LVCMOS33 } [get_ports {s[3]}]
set_property -dict { PACKAGE_PIN W18   IOSTANDARD LVCMOS33 } [get_ports {cout}]
#set_property -dict { PACKAGE_PIN U15   IOSTANDARD LVCMOS33 } [get_ports {led[5]}]
#set_property -dict { PACKAGE_PIN U14   IOSTANDARD LVCMOS33 } [get_ports {led[6]}]
#set_property -dict { PACKAGE_PIN V14   IOSTANDARD LVCMOS33 } [get_ports {led[7]}]
```

```
#set_property -dict { PACKAGE_PIN V13   IOSTANDARD LVCMOS33 } [get_ports {led[8]}]
#set_property -dict { PACKAGE_PIN V3    IOSTANDARD LVCMOS33 } [get_ports {led[9]}]
#set_property -dict { PACKAGE_PIN W3    IOSTANDARD LVCMOS33 } [get_ports {led[10]}]
#set_property -dict { PACKAGE_PIN U3    IOSTANDARD LVCMOS33 } [get_ports {led[11]}]
#set_property -dict { PACKAGE_PIN P3    IOSTANDARD LVCMOS33 } [get_ports {led[12]}]
#set_property -dict { PACKAGE_PIN N3    IOSTANDARD LVCMOS33 } [get_ports {led[13]}]
#set_property -dict { PACKAGE_PIN P1    IOSTANDARD LVCMOS33 } [get_ports {led[14]}]
#set_property -dict { PACKAGE_PIN L1    IOSTANDARD LVCMOS33 } [get_ports {led[15]}]

```

SCREENSHOT OF CODE AND TESTBENCH



```
`timescale 1ns / 1ps
//////////////////////////////////////////////////////////////////////////////////
// Company:
// Engineer:
//
// Create Date: 10/01/2026 04:42:48 PM
// Design Name:
// Module Name: rca4_verilog
// Project Name:
// Target Devices:
// Tool Versions:
// Description:
//
// Dependencies:
//
// Revision:
// Revision 0.01 - File Created
// Additional Comments:
//
    //////////////////////////////////////////////////////////////////////////////////


module rca4_verilog(
        input [3:0] a,b,
        input cin,
        output [3:0] s,
        output cout
    );

    wire [4:0] c;

    assign c[0] = cin;
    assign cout = c[4];

    genvar i;

    generate

        for(i = 0 ; i < 4 ; i = i + 1) begin : fa

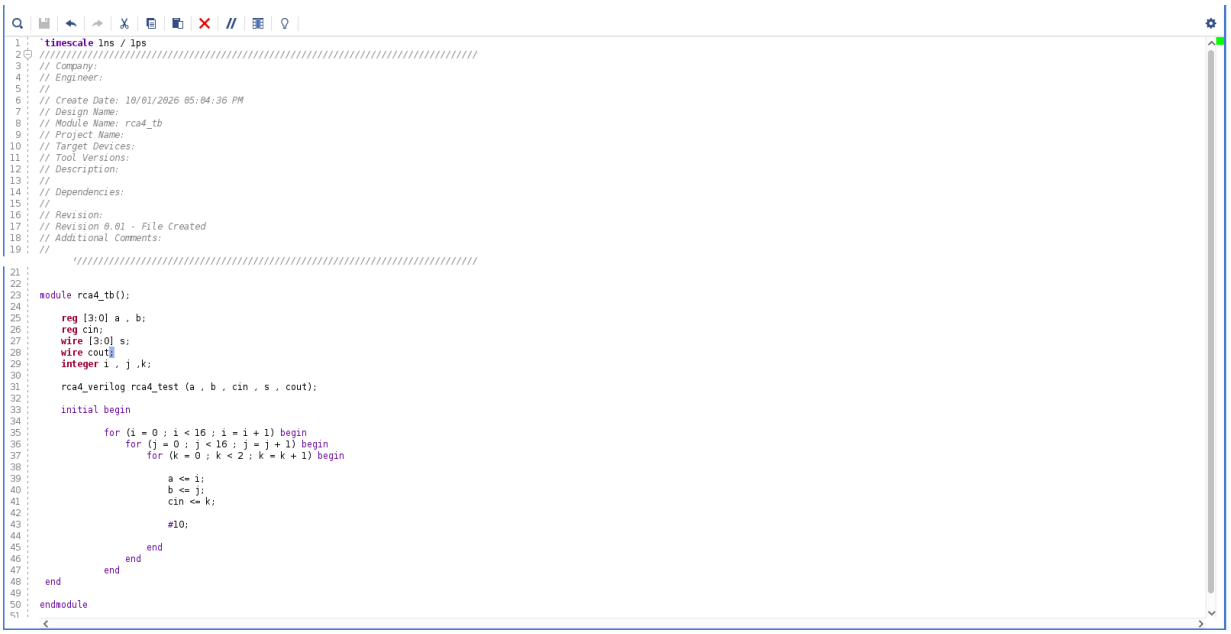
            fa_verilog fa (a[i] , b[i] , c[i] , s[i] , c[i+1]);

        end


    endgenerate

endmodule
```

Figure 1: 4-bit RCA Verilog Design Code



```
`timescale 1ns / 1ps
//////////////////////////////////////////////////////////////////////////////////
// Company:
// Engineer:
//
// Create Date: 10/01/2026 05:04:36 PM
// Design Name:
// Module Name: rca4_tb
// Project Name:
// Target Devices:
// Tool Versions:
// Description:
//
// Dependencies:
//
// Revision:
// Revision 0.01 - File Created
// Additional Comments:
//
    //////////////////////////////////////////////////////////////////////////////////


module rca4_tb();

    reg [3:0] a , b;
    reg cin;
    wire [3:0] s;
    wire cout;
    integer i , j ,k;

    rca4_verilog rca4_test (a , b , cin , s , cout);

    initial begin

            for (i = 0 ; i < 16 ; i = i + 1) begin
                for (j = 0 ; j < 16 ; j = j + 1) begin
                    for (k = 0 ; k < 2 ; k = k + 1) begin

                        a <= i;
                        b <= j;
                        cin <= k;

                        #10;

                    end
                end
            end
 end

endmodule
```

Figure 2: 4-bit RCA Testbench Code

SCREENSHOT OF CONSTRAINTS FILE



```
## This file is a general .xdc for the Basys3 rev B board
## To use it in a project:
## - uncomment the lines corresponding to used pins
## - rename the used ports (in each line, after get_ports) according to the top level signal names in the project

## Clock signal
#set_property -dict { PACKAGE_PIN W5   IOSTANDARD LVCMOS33 } [get_ports clk]
#create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports clk]


## Switches
set_property -dict { PACKAGE_PIN V17   IOSTANDARD LVCMOS33 } [get_ports {a[0]}]
set_property -dict { PACKAGE_PIN V16   IOSTANDARD LVCMOS33 } [get_ports {a[1]}]
set_property -dict { PACKAGE_PIN W16   IOSTANDARD LVCMOS33 } [get_ports {a[2]}]
set_property -dict { PACKAGE_PIN W17   IOSTANDARD LVCMOS33 } [get_ports {a[3]}]
set_property -dict { PACKAGE_PIN W15   IOSTANDARD LVCMOS33 } [get_ports {b[0]}]
set_property -dict { PACKAGE_PIN V15   IOSTANDARD LVCMOS33 } [get_ports {b[1]}]
set_property -dict { PACKAGE_PIN W14   IOSTANDARD LVCMOS33 } [get_ports {b[2]}]
set_property -dict { PACKAGE_PIN W13   IOSTANDARD LVCMOS33 } [get_ports {b[3]}]
        perty -dict { PACKAGE_PIN V2    IOSTANDARD LVCMOS33 } [get_ports {cin}]
#set_property -dict { PACKAGE_PIN T3    IOSTANDARD LVCMOS33 } [get_ports {sw[9]}]
#set_property -dict { PACKAGE_PIN T2    IOSTANDARD LVCMOS33 } [get_ports {sw[10]}]
#set_property -dict { PACKAGE_PIN R3    IOSTANDARD LVCMOS33 } [get_ports {sw[11]}]
#set_property -dict { PACKAGE_PIN W2    IOSTANDARD LVCMOS33 } [get_ports {sw[12]}]
#set_property -dict { PACKAGE_PIN U1    IOSTANDARD LVCMOS33 } [get_ports {sw[13]}]
#set_property -dict { PACKAGE_PIN T1    IOSTANDARD LVCMOS33 } [get_ports {sw[14]}]
#set_property -dict { PACKAGE_PIN R2    IOSTANDARD LVCMOS33 } [get_ports {sw[15]}]


## LEDs
set_property -dict { PACKAGE_PIN U16   IOSTANDARD LVCMOS33 } [get_ports {s[0]}]
set_property -dict { PACKAGE_PIN E19   IOSTANDARD LVCMOS33 } [get_ports {s[1]}]
set_property -dict { PACKAGE_PIN U19   IOSTANDARD LVCMOS33 } [get_ports {s[2]}]
set_property -dict { PACKAGE_PIN V19   IOSTANDARD LVCMOS33 } [get_ports {s[3]}]
set_property -dict { PACKAGE_PIN W18   IOSTANDARD LVCMOS33 } [get_ports {cout}]
#set_property -dict { PACKAGE_PIN U15   IOSTANDARD LVCMOS33 } [get_ports {led[5]}]
#set_property -dict { PACKAGE_PIN U14   IOSTANDARD LVCMOS33 } [get_ports {led[6]}]
#set_property -dict { PACKAGE_PIN V14   IOSTANDARD LVCMOS33 } [get_ports {led[7]}]
#set_property -dict { PACKAGE_PIN V13   IOSTANDARD LVCMOS33 } [get_ports {led[8]}]
#set_property -dict { PACKAGE_PIN V3    IOSTANDARD LVCMOS33 } [get_ports {led[9]}]
#set_property -dict { PACKAGE_PIN W3    IOSTANDARD LVCMOS33 } [get_ports {led[10]}]
#set_property -dict { PACKAGE_PIN U3    IOSTANDARD LVCMOS33 } [get_ports {led[11]}]
#set_property -dict { PACKAGE_PIN P3    IOSTANDARD LVCMOS33 } [get_ports {led[12]}]
#set_property -dict { PACKAGE_PIN N3    IOSTANDARD LVCMOS33 } [get_ports {led[13]}]
#set_property -dict { PACKAGE_PIN P1    IOSTANDARD LVCMOS33 } [get_ports {led[14]}]
#set_property -dict { PACKAGE_PIN L1    IOSTANDARD LVCMOS33 } [get_ports {led[15]}]
```

Figure 3: Basys 3 Constraints File

SIMULATION WAVEFORM



Figure 4: 4-bit RCA Simulation Waveform